

How to give read and write permissions using Unix commands. A very simple example is to create a text file with a C program, then use `chmod`.

Create a file called `permission.c`

```
#include <stdio.h>
int
main()
{
    FILE *fp;    fp =
    fopen("sample.txt", "w");    if
    (fp == NULL)
    {        printf("Cannot open
    file\n");        return 1;
    }    fprintf(fp, "Hello, this is a test
    file.\n");    fclose(fp);    printf("File
    created successfully.\n");    return 0;
}
```

Compile `gcc permission.c -o permission`

Run

`./permission`

Output

File created successfully.

A file called sample.txt is created.

Check the current permissions

```
ls -l sample.txt
```

You may see

```
-rw-r--r-- 1 user user 31 Aug 17 08:20 sample.txt
```

The owner has: Read + Write

Group has: Read Others

have: Read

This is permission 644.

Give Read and Write permission to everyone `chmod`

```
666 sample.txt
```

Check `ls -l sample.txt`

You should see

```
-rw-rw-rw- 1 user user ... sample.txt
```

Instead of numbers, you can directly give:

```
chmod u+rw sample.txt
```

chmod o+rw sample.txt

chmod a+rw sample.txt

Task of the Day

- 1) Give Read and Write only to the owner
- 2) Give Read and Write to the owner and group
- 3) Only owner can read and write
- 4) Owner read/write, everyone else read-only
- 5) Owner can do everything, others can only read and execute
- 6) Owner and group can read/write
- 7) Only owner can execute

Fill in the permission numbers for the following:

- 8) Owner: read + write; Group: nothing; Others: nothing
→ 600
- 9) Owner: read + write; Group: read; Others: read →
644
- 10) Owner: read + write + execute; Group: read + execute;
Others: read + execute → 755
- 11) Owner: read + write; Group: read + write; Others:
nothing → 660
- 12) Everyone: read only → 444

```
v_santosh@DESKTOP-E34M7RI:~$ nano permission.c
v_santosh@DESKTOP-E34M7RI:~$ gcc permission.c -o permission
v_santosh@DESKTOP-E34M7RI:~$ ./permission.c
-bash: ./permission.c: Permission denied
v_santosh@DESKTOP-E34M7RI:~$ ./permission
File created successfully.
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
-rw-r--r-- 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ chmod 666 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
-rw-rw-rw- 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ chmod 600 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
-rw----- 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ chmod 660 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
-rw-rw---- 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ chmod 755 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
-rwxr-xr-x 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ chmod 100 sample.txt
v_santosh@DESKTOP-E34M7RI:~$ ls -l sample.txt
---x----- 1 v_santosh v_santosh 28 Aug 24 05:59 sample.txt
```